

Savitzky-Golay split command-lag search

rotor median publish period [s]: 0.00499773025513
gimbal median publish period [s]: 0.00499868392944
initial lags [s]: {'rotor': 0.004997730255126953, 'gimbal': 0.004998683929443359}
smooth half-width multipliers: [4.0, 2.0, 1.0, 0.5]

smooth width=4: rotor=0.2s gimbal=0.0341817733s cost=1.5011814042 lag_gradient={'rotor':
-0.9398603921131644, 'gimbal': -1.621239824528553e-06} FD=[{'direction':
'negative_gradient', 'scheme': 'backward', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 7.509936553990846, 'finite_difference_l2_norm': 7.509917627559888,
'difference_l2_norm': 0.0007474691650461809, 'relative_error': 9.953068972985787e-05},
{'direction': 'rotor_delay_seconds', 'scheme': 'backward', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 7.509936941527405, 'finite_difference_l2_norm': 7.509918014986703,
'difference_l2_norm': 0.0007474691485614671, 'relative_error': 9.953068239870513e-05},
{'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 15.805263376088954, 'finite_difference_l2_norm': 15.805263268174503,
'difference_l2_norm': 5.981737315848818e-07, 'relative_error': 3.78464893213884e-08}]
smooth width=2: rotor=0.2s gimbal=0.034262264s cost=1.50268348675 lag_gradient={'rotor':
-0.8895068078586029, 'gimbal': -5.2975379030428815e-06} FD=[{'direction':
'negative_gradient', 'scheme': 'backward', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 8.345812541308169, 'finite_difference_l2_norm': 8.34580451439022,
'difference_l2_norm': 0.0028899272721404066, 'relative_error': 0.0003462727275309042},
{'direction': 'rotor_delay_seconds', 'scheme': 'backward', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 8.345815735565038, 'finite_difference_l2_norm': 8.345807708840969,
'difference_l2_norm': 0.0028899272748944514, 'relative_error': 0.0003462725953293281},
{'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 17.34102200374746, 'finite_difference_l2_norm': 17.34102054749901,
'difference_l2_norm': 4.519521121095397e-06, 'relative_error': 2.6062599540665544e-07}]
smooth width=1: rotor=0.2s gimbal=0.0331252046s cost=1.50406865934 lag_gradient={'rotor':
-1.1011738962428195, 'gimbal': 8.947228669980989e-06} FD=[{'direction': 'negative_gradient',
'scheme': 'backward', 'step': 6.0554544523933395e-06, 'analytic_l2_norm':
10.132968506277964, 'finite_difference_l2_norm': 10.133659292993787, 'difference_l2_norm':
0.009362868187664, 'relative_error': 0.0009239375349965934}, {'direction':
'rotor_delay_seconds', 'scheme': 'backward', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 10.132964234009437, 'finite_difference_l2_norm': 10.13365502634131,
'difference_l2_norm': 0.00936286814985336, 'relative_error': 0.0009239379202780857},
{'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 19.286714598165386, 'finite_difference_l2_norm': 19.28670763850207,
'difference_l2_norm': 3.0497152164348893e-05, 'relative_error': 1.5812517994770286e-06}]
smooth width=0.5: rotor=0.2s gimbal=0.0322690168s cost=1.50501030419 lag_gradient={'rotor':
-1.3959237559787903, 'gimbal': 0.0144180491614527} FD=[{'direction': 'negative_gradient',
'scheme': 'backward', 'step': 6.0554544523933395e-06, 'analytic_l2_norm':
12.666926119367604, 'finite_difference_l2_norm': 12.667486897323823, 'difference_l2_norm':
0.029674424060881055, 'relative_error': 0.002342565996034318}, {'direction':
'rotor_delay_seconds', 'scheme': 'backward', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 12.664705633972652, 'finite_difference_l2_norm': 12.665270412761798,
'difference_l2_norm': 0.029677518673013513, 'relative_error': 0.002343220295013189},
{'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 24.075884309874212, 'finite_difference_l2_norm': 23.654196253638265,
'difference_l2_norm': 2.767355704421197, 'relative_error': 0.11494305541608808}]
strict profile iterations: 1
iteration 1: best=(0.2, 0.032269016751) cost=1.50611852719 rotor_range=(0.195002269745, 0.2)
gimbal_range=(0.027270332821, 0.03726770068) expansions=[]

selected rotor lag 0.2 s (40.0182 publish periods)
selected gimbal lag 0.032269016751 s (6.4555 publish periods)
selected objective cost 1.50611811487